ACETIC ACID

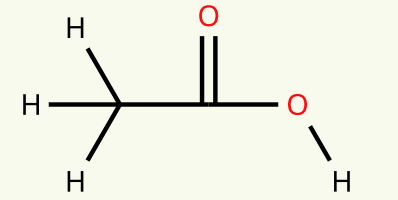
"Simple acid. Diverse potential."



Compound Identity

Common name	**: Acetic acid**
IUPAC name	**: Ethanoic acid**
Molecular formula	**: $C_2H_4O_2$**
molecular weight	**: 60.05 g/mol**

Structure



Structural formula (displayed)

Functional Groups

- Carboxyl group (-COOH)
- Methyl group (-CH3)

Notable feature

• Small carboxylic acid (a carboxylic acid).

Important Properties

Appearance / State
Colorless, clear liquid

Solubility
Miscible with water and most organic solvents.

Boiling Point
118.1 °C

Melting Point
16.6 °C

pKa
~4.76 (Weak acid)

What is Amazing About It?

Acetic acid is the main acid responsible for the sour taste of vinegar. Its –COOH group allows it to donate a hydrogen ion, giving it its acidic properties



It's a foundational chemical for countless materials, from plastics and polymers to medicinal compounds.

Uses / Value

Everyday uses

- Food preservative (vinegar)
- Household cleaner

Medical uses

- Vinyl acetate monomer (for plastics)
- Acetic anhydride (for fibers, textiles)

Industrial uses

- Solvent
- Chemical intermediate

Chemical engineering connec

Produced primarily via carbonylation of methanol and oxidation of acetaldehyde.
A key material for process design.

Risks & Safety

Health risks

- Highly corrosive
- Can irritate the skin and eyes.
- Concentrated acetic acid can be corrosive.
- High concentrations can irritate the respiratory system.
- Irritating vapors; can cause severe skin burns.

Fire / Reactivity

- Flammable liquid and vapor; keep away from heat.

Environmental concerns

- Large releases of concentrated acetic acid should be prevented because they can affect aquatic environments and local ecosystems.

References

National Center for Biotechnology Information (2026). PubChem Compound Summary for CID 176, Acetic Acid. Retrieved September 9, 2026 from National Center for Biotechnology Information. (2025). PubChem compound summary for CID 702, Ethanol. National Center for Biotechnology Information. (n.d.). Ethanol. PubChem. https://pubchem.ncbi.nlm.nih.gov/compound/Ethanol.

Green building block for a sustainable world